

DISCOUNTED CASH FLOW VALUATION

- Prepared for the CFA Institute Investment Foundations Certificate, (Course 3/ Module 2).
- I chose **Freeport-McMoRan**, the 3rd largest copper miner globally.
- I estimate a Value/share of \$86 compared to current market price of \$72 suggesting around 15% undervaluation.
- But the range of outcomes is massive; simulating the copper price inline with historic mean/stdev of 5%/14% pa generates a very wide range of 10-year Free Cash Flow paths, albeit positively skewed.



		1	2	3	4	5	6	7	8	9	10	
USD millions	BASE YEAR	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10	TERMINAL YEAR
DISCOUNTED CASH FLOW	BASE YEAR	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10	TERMINAL YEAR
Net Income	4,152	4,384	4,844	5,482	6,202	7,013	7,924	8,946	10,090	11,370	12,798	14,391
D&A	2,244	2,533	2,631	2,763	2,901	3,046	3,198	3,358	3,526	3,702	3,887	4,081
Income Tax expense	2,221	2,361	2,608	2,952	3,340	3,776	4,267	4,817	5,433	6,122	6,891	7,749
Net Interest	146	146	237	238	221	185	130	53	49	176	332	519
EBITDA	8,763	9,424	10,321	11,434	12,663	14,020	15,519	17,174	19,001	21,018	23,245	25,702
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Capital expenditures	-	4,494	-	4,500	-	4,500	-	4,500	-	4,500	-	4,500
Free Cash Flow from Operations	4,269	4,924	5,821	6,934	8,163	9,520	11,019	12,674	14,501	16,518	18,745	21,202

Discount Rate	10%
cumulated discount factor	
Terminal cash flow	21,202
Terminal cost of capital	10%
Terminal Value	212,024
Present Value (Terminal Value)	81,745
Present Value of next 10 years Cash Flows	59,704
Total Present Value	141,448
Value of Operating assets =	141,448
- Debt	9,379
- Minority Interests	11,867
+ Cash	3,824
Value of Equity	124,026
Number of shares	1,443
Estimated Value / share	85.9
Price on market	72.0
Price as % Value	84%

